



Solid polymer electrolytes based on alternating copolymers of vinyl ethers with methoxy oligo(ethyleneoxy)ethyl groups and vinylene carbonate



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ABSTRACT

Alternating copolymers (poly(1a-g-alt-VC)) of vinyl ethers with various methoxy oligo(ethyleneoxy)ethyl groups and vinylene carbonate (VC) were prepared, and the thermal and electrochemical properties of their polymer electrolytes with LiTFSI and interfacial stability between the polymer electrolyte and Li metal electrode were investigated. T_g 's increased linearly with salt contents, and decreased with an increase in the chain length of methoxy oligo(ethyleneoxy)ethyl groups in the vinyl ethers at constant salt concentration. The slopes of T_g vs. $[\text{Li}]/[\text{O}]$ were identical, independent of the polymer structure. The ionic conductivities of the polymer electrolytes increased with increasing the side-chain ethyleneoxy (EO) unit length of the vinyl ether unit in the alternating copolymers, and also their temperature dependences became relatively smaller in the polymer electrolytes having longer EO units in the vinyl ethers. The highest ionic conductivity, $1.2 \times 10^{-4} \text{ S/cm}$ at 30°C , was obtained in the alternating copolymer with a side-chain EO unit length of 23.5 in the vinyl ether unit. Ion transport coupled with the segmental motion of the polymer is dominant in these polymer electrolytes. Interfacial resistance increased gradually with contact time, indicative of the formation of passivation films on the Li metal electrode. These polymer electrolytes are thermally stable and have large electrochemical windows of use.

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1. Introduction

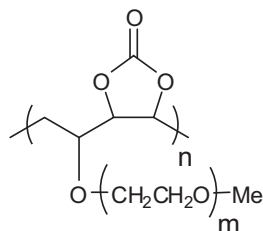
Solid polymer electrolytes have been received attention as electrolyte materials for all solid-state lithium batteries because of advantages in mechanical stability, safety, processability and flexibility in cell design in comparison with the liquid electrolytes and gel-type polymer electrolytes. Poly(ethylene oxide) (PEO) as a typical solid polymer electrolyte has been widely studied, because of good performance, its moderate interaction with lithium ion, high electrochemical stability, easy fabrication, and low cost material [1–6]. However, linear PEO-based polymer electrolytes do not show sufficient ionic conductivity at room temperature regions due to the crystallization of the PEO chains, though they have high ionic conductivity at higher temperature than 60°C . Therefore, numerous attempts to overcome the disadvantage for the PEO-based polymer electrolytes, for example, network formation, usage of random copolymers and block copolymers, addition of

plasticizer such as carbonate- and ether-type solvents, oligomeric PEO, hyperbranched polymers, and ionic fillers, and carbon dioxide treatment have been made [7–11]. It is well-known that lithium ion transport in the PEO-based polymer electrolyte can be realized by the local motion of the EO chain in the amorphous region, that is, strongly coupled to the segmental motion of the polymer [5,6]. When the ionic conduction in the polymer electrolytes takes place in the ion transport mechanism coupled with the segmental mobility, serious decrease in ionic conductivity is unavoidable at lower temperature. On the other hand, for the polymer electrolytes of poly(vinylene carbonate) and poly(1,3-dioxolan-2-one-4,5-diyl oxalate) with lithium salt in high salt concentrations [12], the polymer electrolytes of poly(acrylonitrile) [13] and poly(vinyl alcohol) [14,15] with lithium salt, and the “polymer-in-salt” electrolytes [16], it is reported that the ionic conduction in their electrolytes obeys ion transport mechanism decoupled from the segmental motion. When the ionic conduction in the polymer electrolytes is decoupled to the segmental mobility, temperature dependence of the ionic conductivity for the polymer electrolyte is considered to become relatively smaller. If hybrid-type polymer electrolyte, where both ion transport mechanism coupled with the segmental motion and ion transport one decoupled from the segmental

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motion are combined, is generated, it is expected that the hybrid-type polymer electrolyte has high ionic conductivity over the wide temperature region and also smaller temperature dependence of ionic conductivity. Therefore, we are interested in preparation of the hybrid-type polymer electrolyte and their electrochemical and thermal properties.

In this work, we prepared alternating copolymers (poly(1a-g-alt-VC)) composed of vinyl ethers having various chain length of methoxy oligo(ethyleneoxy)ethyl group, where *m* corresponds to the number of the EO unit, with vinylene carbonate (VC), and investigated the electrochemical and thermal properties of their polymer electrolytes containing a lithium salt, and interfacial stability between the polymer electrolytes and a lithium metal electrode.



Poly(1a-g-alt-VC)
(*m* = 2, 3, 5, 7.5, 10, 15, 23.5)

2. Experimental

2.1. Materials

Methoxy oligo(ethyleneoxy)ethyl vinyl ethers (1e (*m* = 10), 1f (*m* = 15), and 1g (*m* = 23.5)) and diethylene glycol vinyl ether (2) were kindly supplied by Maruzen Petrochemical Co., Ltd. Methyl triethylene glycol (*x* = 3) (Sigma-Aldrich), methoxy (diethyleneoxy)ethyl vinyl ether (1b (*m* = 3)) (Sigma-Aldrich), methoxy oligo(ethylene glycol) (MW = 300, *x* = 5.5) (Sigma-Aldrich), *p*-toluenesulfonyl chloride (TCI Co., Ltd.), sodium hydride (TCI Co., Ltd.), methanol (Wako Pure Chem. Industries, Ltd.) and diethyl ether (Wako Pure Chem. Industries, Ltd.), methyl iodide (Wako Pure Chem. Industries, Ltd.) and commercial anhydrous acetonitrile (Sigma-Aldrich) were used without further purification. Tetrahydrofuran (THF) (bp 65 °C) was distilled over lithium aluminum hydride. 2,2',-Azobis(isobutyronitrile) (AIBN) (Wako Co.) was purified by recrystallization from methanol. Commercial vinylene carbonate (VC) (Kishida Chemical Co., Ltd.) (bp 162 °C) was purified by distillation. Lithium bis(trifluoromethanesulfonimide) (LiTFSI) [LiN(CF₃SO)₂] (Fluka Co.) was dried under vacuum at 120 °C prior to use and kept inside an argon-filled glove box.

2.2. Syntheses of vinyl ethers (1a (*m* = 2), 1c (*m* = 5), 1d (*m* = 7.5))

Vinyl ethers of 1a (*m* = 2), 1c (*m* = 5) and 1d (*m* = 7.5) were prepared by the synthetic routes shown in Scheme 1.

2.2.1. Methoxy(ethyleneoxy)ethyl vinyl ether (1a, *m* = 2)

Sodium hydride (3.2 g, 80 mmol), washed three times with dry THF, was added in dry THF (20 mL), and into the suspension diethylene glycol vinyl ether (2) (9.25 g, 70 mmol) was added dropwisely at room temperature. Into the resulting solution methyl iodide (10.7 g, 75 mmol) was added and then stirred for 1 h. The reaction mixture was placed under reduced pressure to remove the solvent and unreacted methyl iodide, and then the saturated sodium

chloride aqueous solution was added into it. The resulting solution was extracted three times with dichloromethane (100 mL), and the extracts were combined, dried with anhydrous magnesium sulfate. The filtrate was placed under reduced pressure, and then distilled to obtain the methoxy (ethyleneoxy)ethyl vinyl ether (1a, *m* = 2) as colorless oil (bp 73–75 °C/15 mmHg): yield 5.05 g (yield 50%), IR(NaCl) cm⁻¹: ν(CH) 2878, ν(C=C) 1619, ν(C–O) 1111; ¹H-NMR(CDCl₃): δ 6.50(dd, *J* = 6.76, 14.3 Hz, 1H), 4.19(dd, *J* = 2.14, 14.3 Hz, 1H), 4.01(dd, *J* = 2.14, 6.76 Hz, 1H), 3.86(t, *J* = 5.28 Hz, 2H), 3.75(t, *J* = 5.28 Hz, 2H), 3.67(t, *J* = 4.29 Hz, 2H), 3.57(t, *J* = 4.29 Hz, 2H), 3.39(s, 3H); ¹³C-NMR(CDCl₃): δ 151.7(CH₂=CH), 86.6(CH₂=CH), 71.9(CH₂), 70.6(CH₂), 69.6(CH₂), 67.1(CH₂), 59.0(CH₃).

2.2.2. Methoxy (tetraethyleneoxy)ethyl vinyl ether (1c, *m* = 5)

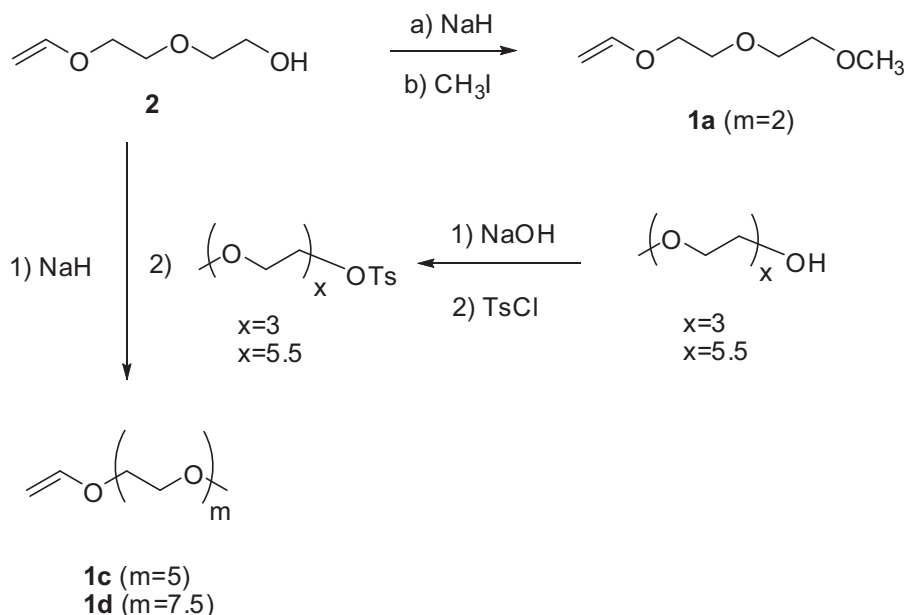
Sodium hydroxide (3.6 g, 90 mmol) was dissolved in 20 mL of distilled water, and into the aqueous solution cooled with ice bath methyl triethylene glycol (10.3 g, 63 mmol) solution in THF (20 mL) was added. Into the resulting solution was added dropwisely the *p*-toluenesulfonyl chloride (12.0 g, 63 mmol) solution in THF (30 mL), and then stirred for 2 h at 0 °C. The reaction mixture was extracted with dichloromethane, and then the extracts was washed twice with water, saturated sodium chloride aqueous solution, and dried with anhydrous magnesium sulfate. The filtrate was placed under reduced pressure to obtain the methyl triethylene glycol tosylate as colorless oil: yield 17.1 g (yield 86%), IR(NaCl) cm⁻¹: ν(CH) 2878, ν(SO₂) 1358, ν(C–O) 1101; ¹H-NMR(CDCl₃): δ 7.80(d, *J* = 8.25 Hz, 2H), 7.34(d, *J* = 8.25 Hz, 2H), 4.16(t, *J* = 4.79 Hz, 2H), 3.69(m, 2H), 3.59(m, 6H), 3.52(m, 2H), 3.37(s, 3H), 2.45(s, 3H); ¹³C-NMR(CDCl₃): δ 144.6(Ar), 132.7(Ar), 129.6(Ar), 127.7(Ar), 71.6(CH₂), 70.5(CH₂), 70.3(CH₂), 69.1(CH₂), 68.4(CH₂), 58.8(CH₃), 21.4(CH₃).

Sodium hydride (3.0 g, 75 mmol), washed three times with dry THF, was added in dry THF (20 mL), and into the suspension diethylene glycol vinyl ether (2) (7.7 g, 58 mmol) was added dropwisely at room temperature. Into the resulting solution methyl diethylene glycol tosylate (18.5 g, 58 mmol) was added and then refluxed for 24 h with stirring. The reaction mixture was placed under reduced pressure to remove the solvent, and then dichloromethane and the saturated sodium chloride aqueous solution were added into it. The organic layer was washed three times with saturated sodium chloride aqueous solution, dried with anhydrous magnesium sulfate. The filtrate was placed under reduced pressure to give crude oil, which was dissolved in a small amount of ethyl acetate. The resulting solution was treated with silica-gel column chromatography (eluent: ethyl acetate) to afford the methoxy (tetraethyleneoxy)ethyl vinyl ether (1c, *m* = 5) as pale yellow oil: yield 2.5 g (yield 15%), IR(NaCl) cm⁻¹: ν(CH) 2873, ν(C=C) 1618, ν(C–O) 1109; ¹H-NMR(CDCl₃): δ 6.50(dd, *J* = 6.93, 14.19 Hz, 1H), 4.19(dd, *J* = 1.65, 14.19 Hz, 1H), 4.02(dd, *J* = 1.65, 6.93 Hz, 1H), 3.85(t, *J* = 4.78 Hz, 2H), 3.74(t, *J* = 4.45 Hz, 2H), 3.64(m, 14H), 3.56(t, *J* = 4.62 Hz, 2H), 3.38(s, 3H); ¹³C-NMR(CDCl₃): δ 151.7(CH₂=CH), 86.6(CH₂=CH), 71.9(CH₂), 70.6(CH₂), 69.6(CH₂), 67.1(CH₂), 59.0(CH₃).

2.2.3. Methoxy oligo(ethyleneoxy)ethyl vinyl ether (1d, *m* = 7.5)

Methyl oligo(ethylene glycol) (*x* = 5.5) tosylate was synthesized as colorless oil (16.6 g, 83% yield) from methyl oligo(ethylene glycol) (MW = 300, *x* = 5.5) (13.2 g, 44 mmol) and *p*-toluenesulfonyl chloride (8.4 g, 44 mmol) by the same method as the procedure of preparing methyl triethylene glycol tosylate: IR(NaCl) cm⁻¹: ν(CH) 2872, ν(SO₂) 1355, ν(C–O) 1103; ¹H-NMR(CDCl₃): δ 7.80(d, *J* = 8.25 Hz, 2H), 7.34(d, *J* = 8.25 Hz, 2H), 4.16(t, *J* = 4.79 Hz, 2H), 3.70–3.53(m, 20H), 3.59(s, 3H), 2.45(s, 3H); ¹³C-NMR(CDCl₃): δ 144.6(Ar), 132.8(Ar), 129.6(Ar), 127.8(Ar), 71.5(CH₂), 70.5(CH₂), 70.0(CH₂), 69.3(CH₂), 68.6(CH₂), 58.8(CH₃), 21.4(CH₃).

Methoxy oligo(ethyleneoxy)ethyl vinyl ether (1d, *m* = 7.5) was synthesized as colorless oil (4.0 g, 26% yield) from methyl



Scheme 1. Synthetic routes of vinyl ethers of 1a ($m=2$), 1c ($m=5$) and 1d ($m=7.5$).

oligo(ethylene glycol) ($x=5.5$) tosylate (16.6 g, 37 mmol) and diethylene glycol vinyl ether (4.8 g, 37 mmol) by the same method as the procedure of preparing methoxy (tetraethyleneoxy)ethyl vinyl ether (1c, $m=5$) as pale yellow oil: IR(NaCl) cm^{-1} : $\nu(\text{CH})$ 2870, $\nu(\text{C}=\text{C})$ 1620, $\nu(\text{C}-\text{O})$ 1110; $^1\text{H-NMR}(\text{CDCl}_3)$: δ 6.50(dd, $J=6.93$, 14.19 Hz, 1H), 4.19(dd, $J=1.65$, 14.19 Hz, 1H), 4.02(dd, $J=1.65$, 6.93 Hz, 1H), 3.84(t, $J=4.78$ Hz, 2H), 3.74(t, $J=4.45$ Hz, 2H), 3.65(m, 24H), 3.56(t, $J=4.62$ Hz, 2H), 3.38(s, 3H); $^{13}\text{C-NMR}(\text{CDCl}_3)$: δ 151.5($\text{CH}_2=\text{CH}$), 86.2($\text{CH}_2=\text{CH}$), 72.8(CH_2), 71.5(CH_2), 70.9(CH_2), 69.2(CH_2), 67.0(CH_2), 58.7(CH_3).

2.3. Copolymerization

Given amounts of vinyl ethers with various chain length of oligo(ethyleneoxy)ethyl groups such as 1a ($m=2$), 1b ($m=3$), 1c ($m=5$), 1d ($m=7.5$), 1e ($m=10$), 1f ($m=15$), or 1g ($m=23.5$), VC as a comonomer, AIBN as a radical initiator were placed in a glass ampule, which was degassed by the freeze-thaw method (repeatedly three times) and sealed. The ampule was placed in a bath thermostated at 60 °C or 90 °C for the time of polymerization and then opened. The reaction mixture was poured into an excess of diethyl ether to precipitate the polymer (poly(1a-g-*alt*-VC)), which was purified in three or more cycles of a redissolution–reprecipitation method. Chloroform and diethyl ether were used as a solvent and precipitant, respectively. The polymers were dried under reduced pressure at room temperature until a constant weight was reached.

Poly(1a($m=2$)-*alt*-VC): Polymer was obtained as white solids. Yield 53%, $M_n=37,000$, IR (NaCl) cm^{-1} : ν 2923(CH), ν 1806(C=O), ν 1105(C–O); $^1\text{H-NMR}(\text{CDCl}_3)$: δ 5.1–4.6(br, 2H), 4.0–3.5(br, 2H), 3.4 (s, 3H), 2.0–1.8(br, 2H); $^{13}\text{C-NMR}(\text{CDCl}_3)$: δ 153.8(C=O), 82.3(CH), 74.8(CH_2), 71.7(CH_2), 70.2(CH), 68.0(CH), 58.7(CH_3), 35.1(CH_2).

Poly(1b($m=3$)-*alt*-VC): Polymer was obtained as pale yellow sticky solid. Yield 70%, $M_n=19,000$, IR (NaCl) cm^{-1} : ν 2876(CH), ν 1805(C=O), ν 1106(C–O); $^1\text{H-NMR}(\text{CDCl}_3)$: δ 5.0–4.8(br, 2H), 4.0–3.5(br, 13H), 3.4 (s, 3H), 2.2–1.8(br, 2H); $^{13}\text{C-NMR}(\text{CDCl}_3)$: δ 153.5(C=O), 82.3(CH), 75.4(CH_2), 71.9(CH_2), 70.5(CH_2), 68.4(CH), 58.7(CH_3), 35.2(CH_2).

Poly(1c($m=5$)-*alt*-VC): Polymer was obtained as pale yellow sticky solid. Yield 79%, $M_n=17,500$, IR (NaCl) cm^{-1} : ν 2873(CH), ν 1806(C=O), ν 1106(C–O); $^1\text{H-NMR}(\text{CDCl}_3)$: δ 5.1–4.4(br, 2H), 4.1–3.5(br, 21H), 3.4(s, 3H), 2.2–1.7(br, 2H); $^{13}\text{C-NMR}(\text{CDCl}_3)$: δ 153.5(C=O), 81.8(CH), 75.4(CH_2), 72.7(CH_2), 70.4(CH_2), 68.4(CH), 58.7(CH_3), 35.2(CH_2).

Poly(1d($m=7.5$)-*alt*-VC): Polymer was obtained as pale yellow sticky solid. Yield 53%, $M_n=7900$, IR (NaCl) cm^{-1} : ν 2871(CH), ν 1806(C=O), ν 1107(C–O); $^1\text{H-NMR}(\text{CDCl}_3)$: δ 5.1–4.7(br, 2H), 4.1–3.5(br, 32.5H), 3.4(s, 3H), 2.2–1.8(br, 2H); $^{13}\text{C-NMR}(\text{CDCl}_3)$: δ 153.5(C=O), 81.9(CH), 75.0(CH_2), 71.8(CH_2), 70.4(CH_2), 68.7(CH), 58.7(CH_3), 35.2(CH_2).

Poly(1e($m=10$)-*alt*-VC): Polymer was obtained as pale yellow sticky solid. Yield 49%, $M_n=8100$, IR (NaCl) cm^{-1} : ν 2860(CH), ν 1810(C=O), ν 1112(C–O); $^1\text{H-NMR}(\text{CDCl}_3)$: δ 4.8–4.6(br, 2H), 3.65–3.45(br, 41H), 3.37 (s, 3H), 1.70–1.60(br, 2H); $^{13}\text{C-NMR}(\text{CDCl}_3)$: δ 153.6(C=O), 81.6(CH), 74.4(CH_2), 71.7(CH_2), 70.7(CH_2), 69.9(CH), 58.7(CH_3), 36.1(CH_2).

Poly(1f($m=15$)-*alt*-VC): Polymer was obtained as pale yellow sticky solid. Yield 71%, $M_n=7800$, IR (NaCl) cm^{-1} : ν 2876(CH), ν 1805(C=O), ν 1106(C–O); $^1\text{H-NMR}(\text{CDCl}_3)$: δ 4.8–4.5(br, 2H), 3.68–3.54(br, 61H), 3.38 (s, 3H), 1.55–1.38(br, 2H); $^{13}\text{C-NMR}(\text{CDCl}_3)$: δ 153.4(C=O), 81.8(CH), 77.4(CH_2), 71.5(CH_2), 70.9(CH_2), 61.1(CH), 58.6(CH_3), 34.9(CH_2).

Poly(1g($m=23.5$)-*alt*-VC): Polymer was obtained as pale yellow sticky solid. Yield 30%, $M_n=7400$, IR (NaCl) cm^{-1} : ν 2871(CH), ν 1803(C=O), ν 1107(C–O); $^1\text{H-NMR}(\text{CDCl}_3)$: δ 5.1–4.8(br, 2H), 3.71–3.53(br, 95H), 3.37 (s, 3H), 1.62–1.60(br, 2H); $^{13}\text{C-NMR}(\text{CDCl}_3)$: δ 151.5(C=O), 86.6(CH), 77.4(CH_2), 71.8(CH_2), 70.4(CH_2), 68.7(CH), 58.8(CH_3), 30.4(CH_2).

2.4. Preparation of polymer electrolytes

All preparation procedure carried out inside a dry argon-filled glove box kept at dew point of –85 °C to avoid moisture contamination. Given amounts of alternating polymers (poly(1a-g)-*alt*-VC) and an appropriate amount of a lithium salt (LiTFSI) were dissolved in acetonitrile. The viscous solution was poured on a Teflon petri dish and acetonitrile was evaporated very slowly at room

temperature to prepare a casting film. Finally, the film was dried at 90 °C for 24 h using electric furnace equipped in the glove box. The obtained polymer electrolyte was peeled from the Teflon laboratory dish and stored inside the glove box.

2.5. Measurements

The ^1H - and ^{13}C -NMR and IR spectra were recorded on JEOL JNM-EX270 FT NMR spectrometer using chloroform-*d* with tetramethylsilane as an internal standard at 25 °C and JASCO IR-700 spectrometer, respectively. The number-average molecular weights (M_n) of the polymers were determined by gel permeation chromatography (GPC) (Tosoh HLC-803D with two columns of Tosoh TSKgel Multipore H_{XL}-M) with standard polystyrenes as a reference and THF as an eluent.

The ionic conductivities of the polymer electrolytes were measured by the complex impedance spectroscopy. The samples were fixed inside a Teflon O-ring spacer with known thickness and sandwiched between two stainless steel (SS) electrode discs acting as ion-blocking electrodes and set in a thermostat oven chamber. The measurements were carried out using Solartron 1260 frequency response analyzer over a frequency range of 1 Hz to 1 MHz and in a temperature range of –20–130 °C with an amplitude of 10 mV. All samples were first kept at 80 °C for at least 12 h and then measured by heating cycle. The measurements were carried out after keeping the samples for 1 h at each temperature to attain thermal equilibration. Nyquist plots were computed from the raw experimental data. The resistance of the membrane is taken at the maximum impedance value. The conductivity values (σ) were derived from the equation of $\sigma = (1/R_b)(L/A)$, where R_b is the bulk resistance determined from the Nyquist plot, L is the thickness of the sample, and A is the area of the sample.

The electrochemical stability (oxidation stability) of the polymer electrolytes was evaluated using Solartron 1287 electrochemical interface by running a linear sweep voltammetry (10 mV/s) at room temperature in a Li/polymer electrolyte/SS cell, where Li and SS were used as a counter electrode and a blocking working electrode, respectively.

Lithium ion transport number (T_+) was determined by the two-impedance polarization coupling technique developed by Bruce et al., where cells with symmetrical non-blocking lithium metal electrode were used [17,18].

The thermal stability of the polymer electrolyte was investigated with thermogravimetry-differential thermal analysis instrument TG-DTA 6200 (Seiko Instruments Inc.) at a heating rate of 5 °C/min in air. The glass transition temperature (T_g) of the polymer electrolytes were determined by the differential scanning calorimetry (DSC) using EXSTER6000 thermal analysis instrument DSC 6200 (Seiko Instruments Inc.) under a nitrogen gas flow. About 10 mg amount of sample was weighted, loaded in an aluminum pan, and then sealed. The measurement was carried out in a temperature range of –100–110 °C at a heating rate of 10 °C/min.

3. Results and discussion

3.1. Synthesis of copolymers

The copolymerizations of vinyl ethers such as 1a ($m=2$), 1b ($m=3$), 1c ($m=5$), 1d ($m=7.5$), 1e ($m=10$), 1f ($m=15$), and 1g ($m=23.5$) with VC at an equimolar feed in bulk were carried out in the presence of AIBN at 60 or 90 °C. The results are summarized in Table 1.

The polymers were obtained as white solids for 1a($m=2$) to pale yellow sticky solids for 1b-g, and their number-average molecular weights (M_n) and molecular weight distributions (M_w/M_n) were in the range of 7400–37,000, and in the range of 1.2–3.2, respectively. IR spectra of all polymers showed the absorption peaks at 1803–1810 cm^{-1} due to a carbon–oxygen double-bond stretching vibration of a carbonyl group and at 1105–1107 cm^{-1} due to a carbon–oxygen single-bond stretching vibration of an ether group, indicating that all polymers contain both vinyl ether and VC units. The peak area ratios in the ^1H -NMR spectra for all polymers were in good agreement with the calculated ratios for the alternating copolymers of vinyl ethers with VC. And also, each peak in the ^{13}C -NMR spectra for all polymers was reasonably assigned to each alternating copolymer of the corresponding vinyl ether with VC. It is, therefore, concluded that all polymers obtained by these copolymerizations are to be alternating copolymers (Scheme 2). It is reported that the copolymerization of methoxy (diethylenoxy)ethyl vinyl ether with VC affords a corresponding alternating copolymer [19]. This strongly supports the formation of alternating copolymers (poly(1a-g-alt-VC)) by the copolymerizations of vinyl ethers and VC in this work.

3.2. Thermal properties of the alternating copolymers and their polymer electrolytes

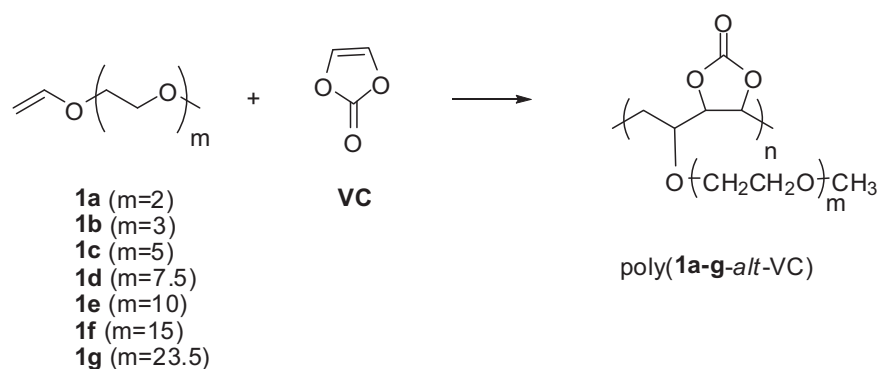
Thermal properties such as glass transition temperature (T_g), melting temperature (T_m), and decomposition temperatures (T_{d5}) at 5 wt% loss of all alternating copolymers (poly(1a-g-alt-VC)) obtained by the copolymerizations of vinyl ethers with VC and of poly(1a-g-alt-VC)/LiTFSI electrolytes at the $[\text{Li}]/[\text{O}] = 1/16$ were measured by TG-DTA and DSC measurements, and the results are summarized in Table 2 and Fig. 1.

The T_g values of the poly(1a-g-alt-VC) decreased with an increase in the side-chain EO unit length in the alternating copolymers, and increased slightly at the longer side-chain EO unit length of 15 and 23.5. Melting points (T_m) of the alternating copolymers except for poly(1d-alt-VC) were observed in the range of 11–42 °C, and they increase with increasing the side-chain EO unit length, approaching to the crystallization temperature of linear PEO. This indicates that crystallization of the side-chain EO unit of the alternating copolymers takes place, and also thermal behavior of the alternating copolymers is highly dependent upon the monomer structure. Alternating copolymers (poly(1a-g-alt-VC)) have the T_{d5} of about 300 °C, indicating that they are thermally stable. As shown

Table 1
Bulk copolymerizations of vinyl ethers such as 1a($m=2$), 1b($m=3$), 1c($m=5$), 1d($m=7.5$), 1e($m=10$), 1f($m=15$), and 1g($m=23.5$) with vinylene carbonate (VC) in the presence of AIBN.

Run	1a-g (g)	VC (g)	[1a-g + VC]/AIBN	Time (h)	Yield (%)	M_n	M_w/M_n
1	1a ($m=2$) 2.0	1.18	200	8	54	37,000	2.2
2	1b ($m=3$) 3.5	1.90	150	18	70	19,000	3.0
3	1c ($m=5$) 2.5	0.78	50	18	79	18,000	3.2
4	1d ($m=7.5$) 2.0	0.44	10	18	53	7900	1.6
5	1e ($m=10$) 1.0	0.17	40	18	49	8100	1.4
6	1f ($m=15$) 2.0	0.24	10	18	72	7800	1.5
7	1g ($m=23.5$) 2.1	0.16	10	18	31	7400	1.2

Polymerization temperature: 60 °C for runs 1–4 and 90 °C for runs 5–7.



Scheme 2. Synthesis of alternating copolymers by copolymerizations of vinyl ethers (1a-g) with VC.

Table 2

Glass transition temperatures (T_g), melting temperatures (T_m), and decomposition temperatures at 5 wt% loss for the poly(1a-g-alt-VC) and poly(1a-g-alt-VC)/LiTFSI electrolytes at the [Li]/[O] of 1/16.

	Alternating copolymers poly(1a-g-alt-VC)			Poly(1a-g-alt-VC)/LiTFSI electrolytes at [Li]/[O] = 1/16		
	T_g (°C)	T_m (°C)	T_{d5} (°C)	T_g (°C)	T_m (°C)	T_{d5} (°C)
1a ($m=2$)	11	20	285	18	nd	300
1b ($m=3$)	−12	20	318	8	nd	308
1c ($m=5$)	−31	11	280	−16	nd	299
1d ($m=7.5$)	−52	nd	330	−27	nd	307
1e ($m=10$)	−60	12	322	−34	nd	321
1f ($m=15$)	−58	24	334	−41	nd	326
1g ($m=23.5$)	−55	42	339	−44	nd	344

nd: not detected.

in Table 2, for all of the poly(1a-g-alt-VC)/LiTFSI electrolytes the T_g rises by the complexation of the poly(1a-g-alt-VC) with LiTFSI. Furthermore, the T_g increases linearly with increasing salt concentration (Fig. 1). This is due to the cross-linking formation by the coordination of oxygen atoms in the polymers to lithium cations [20], which limits the mobility of the polymer chain, resulting in a higher T_g . The slopes of T_g vs. [Li]/[O] are identical, independent of the polymer structure. And also, T_g 's of the poly(1a-g-alt-VC)/LiTFSI electrolytes at the [Li]/[O] = 1/16 decreased with increasing the side-chain EO unit length in alternating copolymers, and the lowest T_g was observed in the polymer electrolyte of the alternating copolymer (poly(1g-alt-VC)) with the side-chain EO unit length of 23.5. The T_{d5} of all polymer electrolytes were observed in the range of 299–344°C, and also T_{d5} values of the polymer electrolytes increased with increasing the side-chain EO unit length

in the alternating copolymers when they were compared with at the same [Li]/[O] ratios. The highest T_{d5} value was found for the poly(1g($m=23.5$)-alt-VC)/LiTFSI electrolyte with the longest side-chain EO unit length. This indicates that the presence of the side-chain EO unit in the polymer structure might affect to the thermal stability of the polymer electrolytes as shown in Table 2. Anyway, these polymer electrolytes investigated in this work have suitable thermal stability as electrolytes for the lithium polymer battery application.

3.3. Ionic conductivity of the polymer electrolytes

Ionic conductivities of the polymer electrolytes composed of the alternating copolymers (poly(1a-g-alt-VC)) with LiTFSI were measured as a function of the temperature and the salt concentrations. Ionic conductivities of these polymer electrolytes at the four different [Li]/[O] ratios of 1/8, 1/12, 1/16 and 1/20 in the temperature range of −20°C to 130°C were shown in Fig. 2a for the poly(1a($m=2$)-alt-VC), Fig. 2b for the poly(1b($m=3$)-alt-VC), Fig. 2c for the poly(1c($m=5$)-alt-VC), Fig. 2d for the poly(1d($m=7.5$)-alt-VC), Fig. 2e for the poly(1e($m=10$)-alt-VC), Fig. 2f for the poly(1f($m=15$)-alt-VC), and Fig. 2g for the poly(1g($m=23.5$)-alt-VC), respectively.

In each case, the ionic conductivity increases with increasing temperature as is typical of PEO-based amorphous polymer electrolytes, which showed the convex curves in the 20°C to 80°C [21]. The ionic conductivities of the polymer electrolytes were well interpreted using the Vogel–Tamman–Fulcher (VTF) [22–24]. All of the polymer electrolytes showed relatively high ionic conductivity at low salt concentrations such as the [Li]/[O] ratios of 1/16 and 1/20. Among all of these polymer electrolytes, the highest ionic conductivity, 1.2×10^{-4} S/cm at 30°C, was observed in the polymer electrolyte containing an alternating copolymer (poly(1g-alt-VC)) with the side-chain EO unit length of 23.5 in the vinyl ether unit. In order to investigate the effect of the side-chain EO unit length of the vinyl ether unit in the polymer structure on the

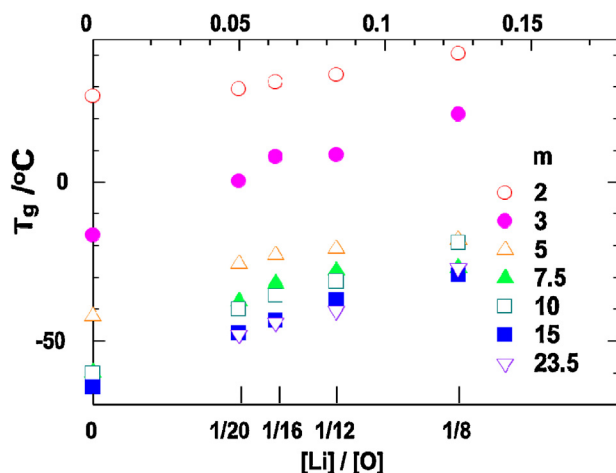


Fig. 1. T_g of the poly(1a-g-alt-VC) and the polymer electrolytes composed of the poly(1a-g-alt-VC) with LiTFSI as a function of LiTFSI concentration.

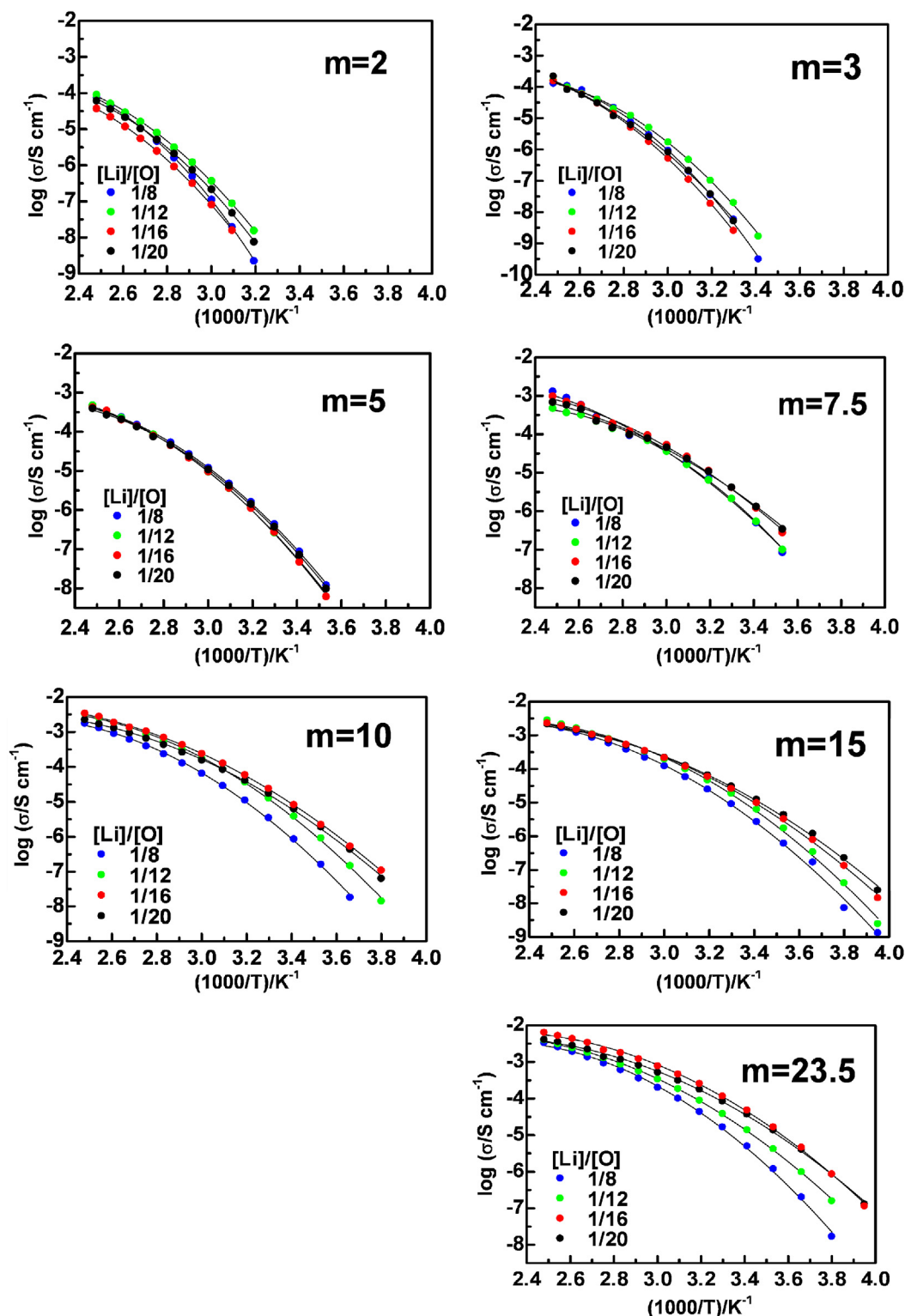


Fig. 2. Temperature dependence of the ionic conductivity for the poly(1a-g-alt-VC)/LiTFSI electrolytes such as 1a($m=2$), 1b($m=3$), 1c($m=5$), 1d($m=7.5$), 1e($m=10$), 1f($m=15$), and 1g($m=23.5$) at four different $[Li]/[O]$ ratios of 1/8, 1/12, 1/16 and 1/20.

ionic conductivity, ionic conductivities were compared among the poly(1a-g-alt-VC)/LiTFSI electrolytes at the $[Li]/[O]$ ratio of 1/16. The results were shown in Fig. 3a, and also the relationship of their ionic conductivities with the side-chain EO unit length of the vinyl ether unit at selected three different temperatures of 30 °C, 80 °C, and 130 °C was shown in Fig. 3b.

The ionic conductivities of the polymer electrolytes increase with an increase in the side-chain EO unit length of the vinyl ether unit in the polymer structure, and also temperature dependence of the ionic conductivities became relatively smaller with increasing the side-chain EO unit length of the vinyl ethers. The poly(1g-alt-VC)/LiTFSI electrolyte with the side-chain EO unit length of 23.5

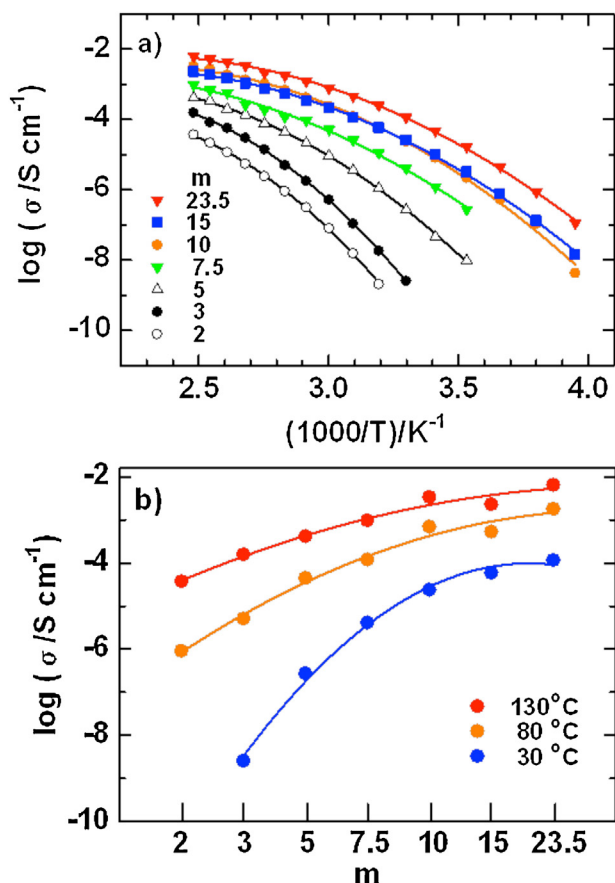


Fig. 3. (a) Temperature dependence of the ionic conductivity for the various poly(1a-g-alt-VC)/LiTFSI electrolytes at the [Li]/[O] ratio of 1/16. (b) The relationship of the ionic conductivities at 30, 80, and 130 °C for the various poly(1a-g-alt-VC)/LiTFSI electrolytes at the [Li]/[O] ratio of 1/16 with a side-chain EO unit length (m) in the vinyl ethers.

in the vinyl ether unit showed the highest ionic conductivity in the wide temperature range. Similar behavior was observed in poly(*p*-phenylene)s containing oligo(ethyleneoxy) group [25] and polycarbonate containing oligo(ethyleneoxy) group [26]. This indicates that the ionic conductivity of these polymer electrolytes is dominated by segmental motion.

3.4. Electrochemical properties of the polymer electrolytes

Electrochemical stability (oxidation stability) of the poly(1a-g-alt-VC)/LiTFSI electrolytes was evaluated by measuring the breakdown voltage by a linear sweep voltammetry, and also the transference number of the lithium ion at 80 °C for the poly(1e-g-alt-VC)/LiTFSI electrolytes and 100 °C for the poly(1b-d-alt-VC)/LiTFSI electrolytes at the [Li]/[O] ratio of 1/16 were determined by the two-impedance polarization coupling technique, and the results were summarized in Table 3.

The breakdown voltages for all of polymer electrolytes at 80 °C were found to be 4.1–4.6 V, which are similar to the value of PEO-based polymer electrolyte. The transference number (T_+) of lithium ion at 80 °C for the polymer electrolytes except for poly(1a(m =2)-alt-VC) were to be 0.11–0.3, similar to those (0.1–0.3) of conventional PEO-based polymer electrolytes [6]. The T_+ values do not change significantly with the side-chain EO unit length in the alternating copolymers. It is well-known that the ion transport in the conventional PEO-based polymer electrolyte

Table 3

Breakdown voltages and transference number for the poly(1a-g-alt-VC)/LiTFSI electrolytes at the [Li]/[O] of 1/16.

run	Poly(1a-g-alt-VC)/LiTFSI electrolytes	Breakdown voltage at 80 °C/V	T_+
1	1a (m =2)	4.1	–
2	1b (m =3)	4.2	0.30
3	1c (m =5)	4.5	0.11
4	1d (m =7.5)	4.6	0.17
5	1e (m =10)	4.1	0.15
6	1f (m =15)	4.3	0.20
7	1g (m =23.5)	4.6	0.16

Measured at 100 °C for runs 2–4 and 80 °C for runs 5–7.

is coupled with segmental motions. Ion transport in the poly(1a-g-alt-VC)/LiTFSI electrolytes is considered to be mainly coupled with the segmental motion of the side-chain EO unit.

3.5. Interfacial property

Compatibility of the polymer electrolytes with electrode materials is an important factor for the polymer battery applications. In order to evaluate interfacial stability between a lithium metal and the poly(1g(m =23.5)-alt-VC)/LiTFSI electrolyte with the [Li]/[O] ratio of 1/12 at 30 °C, the time dependence of interfacial resistance was measured by using the Li metal/poly(1g(m =23.5)-alt-VC)/LiTFSI electrolyte/Li metal cell, and monitored for 1512 h (63 days). The typical Cole–Cole plots of the cell at given times were shown in Fig. 4.

Two semicircles, a high-frequency semicircle and a low-frequency one, were observed. The high-frequency semicircle agreed well with that obtained when blocking stainless steel electrodes were used. Therefore, first, a high-frequency semicircle is assigned to bulk resistance (R_b) of the polymer electrolyte and a low-frequency one is assigned to interfacial resistance (R_i) between a Li metal and the polymer electrolyte, respectively. Time dependence of the R_b and R_i was shown in Fig. 5.

The R_b values are almost to be constant for a long time. On the other hand, R_i values gradually increase with contact time, but the increase in the interfacial resistance value is only twice even at 63 days later. This suggests that passivation films, so-called solid electrolyte interface (SEI), is formed on lithium electrode in the initial stage, and the SEI film formed on the Li metal in contact with the poly(1g(m =23.5)-alt-VC)/LiTFSI electrolyte is relatively stable. It is reported that a stable interface, SEI film, is formed between PEO-based cross-linked polymer electrolyte containing LiTFSI and

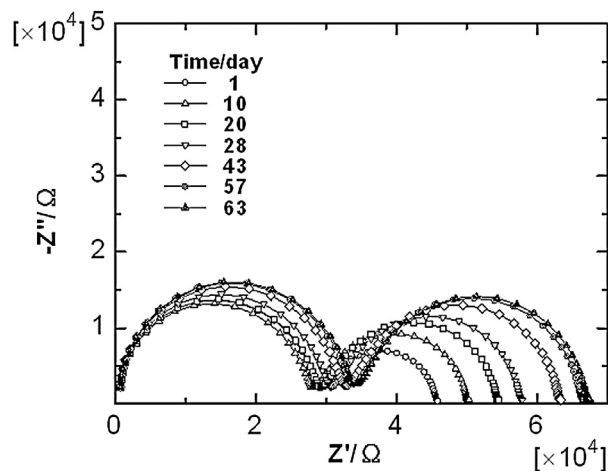


Fig. 4. Nyquist plots of the impedance response of the Li/poly(1g(m =23.5)-alt-VC)/LiTFSI electrolyte/Li cell at 30 °C at 1, 10, 20, 28, 43, 57, and 63 days.

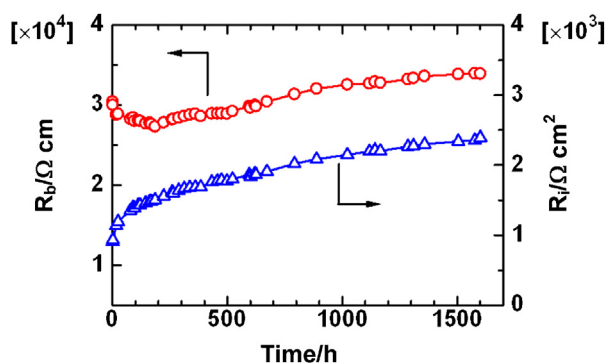


Fig. 5. Time dependence of the bulk resistance (R_b) (○) and the interfacial resistance (R_i) (△) at 30 °C for the Li/poly(1g-alt-VC)/LiTFSI electrolyte/Li cell.

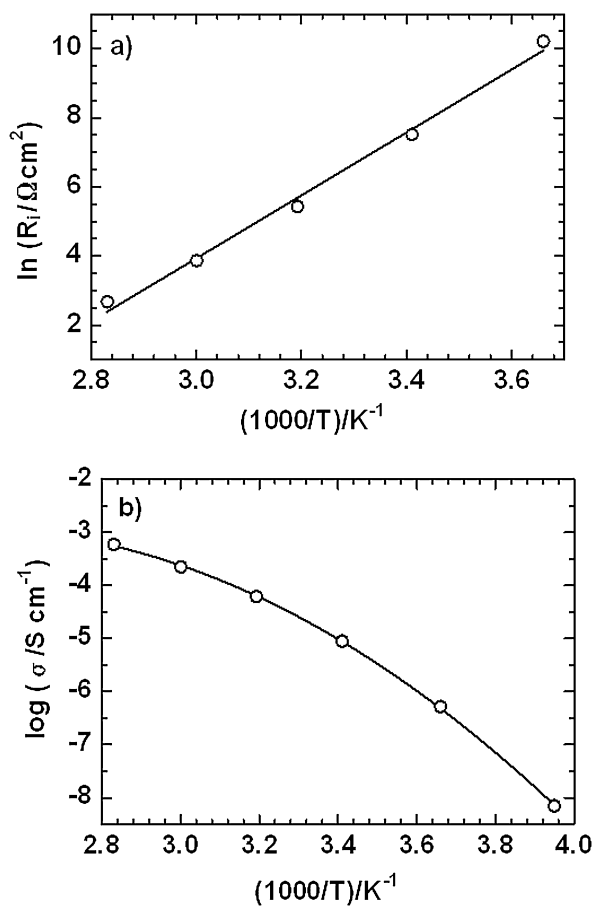


Fig. 6. Arrhenius plot of (a) interfacial resistance (R_i) and (b) ionic conductivity (σ) at the Li metal electrode for the poly(1g($m=23.5$)-alt-VC)/LiTFSI electrolyte with the [Li]/[O] ratio of 1/12.

Li metal, and LiTFSI has a greater effect in the formation of the passivation layer [27].

Arrhenius plot of the interfacial resistance (R_i) at the Li metal interface and the ionic conductivity, σ , for the poly(1g($m=23.5$)-alt-VC)/LiTFSI electrolyte with the [Li]/[O] ratio of 1/12 was shown in Fig. 6a and b.

The Arrhenius plot of R_i gives a straight line, indicating that the interfacial reaction is thermally activated processes, whereas that of σ gives a convex curve, which fits VTF equation. R_i at the Li interface decreases with increasing the ionic conductivity of the polymer electrolyte. The activation energy was calculated from the slope of the Arrhenius plot to be 75.7 kJ/mol, which is little larger than those

of R_i values (57–68 kJ/mol) at the Li metal electrode interface for the copolymer of ethylene oxide with 2-(2-methoxyethoxy)ethyl glycidyl ether containing LiTFSI [28] and of R_i value (67.3 kJ/mol) for mono-acrylate and tri-acrylate of copolymers of ethylene oxide and propylene oxide containing LiTFSI [27]. And also, it is reported that, on the basis of the systematic characterization of electrokinetics of the Li/Li⁺ couple in the poly(ethylene glycol dimethyl ether) (PEGM) and 1,2-dimethoxyethane using LiAsF₆, the exchange current density in PEGM is strongly temperature dependent with a activation energy of 66 kJ/mol, which is characterized to the electron-transfer reaction [29]. It is considered, therefore, that higher activation energy observed in the poly(1g($m=23.5$)-alt-VC)/LiTFSI electrolyte compared to the those of other PEO-based polymer electrolytes comes from the difference in the polymer structure between poly(1g($m=23.5$)-alt-VC) and other PEO-based polymers, where poly(1g($m=23.5$)-alt-VC) contains the ethylene carbonate unit as more polar structure, showing stronger interaction to the Li⁺, in comparison with ether one.

4. Conclusions

Alternating copolymers (poly(1a-g-alt-VC)) were obtained successfully by the copolymerizations of vinyl ethers with various methoxy oligo(ethyleneoxy)ethyl groups and vinylene carbonate (VC) in bulk in the presence of AIBN. Alternating copolymers having long side chains such as poly(1e-g-alt-VC) showed the melting points due to the crystallization of long EO chain. The complexation of poly(1a-g-alt-VC) with LiTFSI formed amorphous polymer electrolytes. Thermal and electrochemical properties of poly(1a-g-alt-VC)/LiTFSI electrolytes and interfacial stability between their polymer electrolytes and Li metal electrode were investigated. The T_g 's increased linearly with LiTFSI contents, and decreased with an increase in the side-chain length of vinyl ether unit at the constant [Li]/[O] ratio. The ionic conductivities of the polymer electrolytes increased with increasing side-chain length of the vinyl ether unit in the alternating copolymers, and also their temperature dependences became relatively smaller in the polymer electrolytes having much longer EO unit. The highest ionic conductivity, 1.2×10^{-4} S/cm at 30 °C, was obtained in the polymer electrolyte with a side-chain EO unit length of 23.5 in the vinyl ether unit. In these polymer electrolytes, ion transport coupled with the segmental motion of the polymer was dominant. These polymer electrolytes are thermally stable and have large electrochemical windows of use. Interfacial resistance increased gradually with time, suggestive of the formation of passivation films on the Li electrode.

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